



MATHEMATICS

Preliminary Course Assessment Task 1

March 8, 2013

General instructions

- Working time – 50 minutes.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the writing paper used in the correct order within this paper and hand to examination supervisors.
- Commence each new question on a new page. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

NAME: # SHEETS USED:

Class (please ✓)

☐ 11M2A – Mr Lin

Marker's use only.

QUESTION(S)	1	2	3	4	5	6	Total	%
MARKS	$\overline{7}$	$\overline{20}$	$\overline{15}$	$\overline{7}$	$\overline{8}$	$\overline{13}$	$\overline{70}$	

Question 1 (Commence a new page)

- a) (i) Evaluate $\sqrt[3]{23.6}$ correct to one decimal place. 1
- (ii) Express 3×10^{-5} as a fraction. 1
- b) Express 346.705529 correct to
- (i) 3 decimal places. 1
- (ii) 3 significant figures. 1
- c) Voyager Spaceship has travelled 75 960 000km. 1
Express this distance in scientific notation.
- d) Express the following as common fractions in their simplest form.
Show all necessary working.
- (i) $0.\dot{6}$ 1
- (ii) $0.58\dot{3}$ 1

Question 2 (Commence a new page)

- a) Expand and simplify
- (i) $4(2t - 1) + t(3t + 2) - t(3t^2 - 2)$ 2
- (ii) $(7r + 2)(7r - 2)$ 1
- (iii) $(3x + 4)^2 - (2x - 5)^2$ 2

b) Factorise completely

(i) $4a + 4b - ax - bx$ **2**

(ii) $72 - 2x^2$ **2**

(iii) $5k^3 + 40$ **1**

(iv) $2x^2 + 11x + 12$ **2**

c) Simplify

(i) $\frac{a}{a-b} + \frac{b}{b-a}$ **2**

(ii) $\frac{x^3 - 1}{x^2 - 1} \div \frac{3x^2 + 3x + 3}{x^2 - 4x - 5}$ **3**

d) Simplify $\frac{1}{x^2 - 4} + \frac{1}{x + 2} - \frac{5}{3x - 6}$ **3**

Question 3 (Commence a new page)

a) Simplify:

(i) $\sqrt{18} - \sqrt{50} - \sqrt{72}$ **2**

(ii) $(\sqrt{3} - 2)(4\sqrt{3} - 6)$ **2**

(iii) $9\sqrt{6} \times 7\sqrt{6}$ **1**

(v) $(\sqrt{5} - 3\sqrt{2})^2$ **2**

b) Express as a surd with a rational denominator.

(i) $\frac{\sqrt{5} - 2}{3\sqrt{5}}$ **2**

(ii) $\frac{2\sqrt{5} - \sqrt{2}}{3\sqrt{5} + 2\sqrt{2}}$ **3**

c) Given $x = \sqrt{5} + 2$, find b if $x + \frac{1}{x} = 2\sqrt{b}$ **3**

Question 4 (Commence a new page)

a) Solve:

(i) $\frac{x+6}{5} = 3$ 1

(ii) $\frac{x+5}{2} - \frac{x-1}{3} = 1$ 2

(iii) $2(3x+7) = 6 - (x-1)$ 2

b) The Fahrenheit temperature Suzie need to cook at is 383°F . 2
However she has an oven with Celsius range only.

Using the formula $F = \frac{9C}{5} + 32$ (where F = degrees Fahrenheit and C = degrees Celsius), Suzie can convert to Celsius degrees.

What temperature in Celsius will she cook at?

Question 5 (Commence a new page)

a) Solve the following for x

(i) $|x-3| = 5$ 2

(ii) $|2x+6| = 3x-1$ 3

b) Solve the following equations simultaneously: 3

$2x + 3y + 1 = 0$ and $x + 2y = 4$

Question 6 (Commence a new page)

- a) Solve
- (i) $x^2 - 5x - 6 = 0$ **1**
 - (ii) $3x^2 - 14x + 8 = 0$ **2**
 - (iii) $3x^2 - 4x - 1 = 0$ (leave answers in exact surd form) **2**
- b) Eight more than twice a number is five less than the number itself.
- (i) Write an algebraic expression for this statement. **1**
 - (ii) What is the number that satisfies this statement? **1**
- c) Solve $3 \leq 2(6 + x)$ and graph its solution on a number line. **3**
- d) Solve $x^2 - x - 30 < 0$ **2**
- e) Solve $|4x + 10| > 13$ **2**

End of examination

Question 1

- (a) (i) 2.9 1
- (ii) $\frac{3}{100000}$ 1
- (b) (i) 346.706 1
- (ii) 347 1
- (c) 7.596×10^7 1
- (d) (i) $\frac{6}{9}$ 1
- (ii) $\frac{7}{12}$ 1

Question 2

- (a) (i) $8t - 4 + 3t^2 + 2t - 3t^3 + 2t$ 1 (correct expansion)
- $= -3t^3 + 3t^2 + 12t - 4$ 1 (correct answer)
- (ii) $49r^2 - 4$ 1
- (iii) $9x^2 + 24x + 16 - 4x^2 + 20x - 25$ 1 (correct expansion)
- $= 5x^2 + 44x - 9$ 1 (correct answer)
- (b) (i) $4(a + b) - x(a + b)$ 1 (attempt at using 'grouping in pairs' method)
- $= (a + b)(4 - x)$ 1
- (ii) $2(36 - x^2)$ 1
- $2(6 + x)(6 - x)$ 1
- (iii) $5(k + 2)(k^2 - 2k + 4)$ 1
- (iv) $(2x + 3)(x + 4)$ 1
- (c) (i) $\frac{a-b}{a-b}$ 1
- $= 1$ 1
-
- (ii) $\frac{(x-1)(x^2+x+1)}{(x+1)(x-1)} \times \frac{(x-5)(x+1)}{3(x^2+x+1)}$ 2 (-1 for each mistake)
- $= \frac{x-5}{3}$ 1 (correct answer)
- (d) $\frac{3+3(x-2)-5(x+1)}{3(x+2)(x-2)}$ 1
- $\frac{3+3x-6-5x-5}{3(x+2)(x-2)}$ 1
- $\frac{-2x-8}{3(x+2)(x-2)}$ 1

Question 3

- | | | |
|---------|--|-------------------------|
| (a) (i) | $3\sqrt{2} - 5\sqrt{2} - 6\sqrt{2}$ | 1 |
| | $= -8\sqrt{2}$ | 1 |
| (ii) | $12 - 6\sqrt{3} - 8\sqrt{3} + 12$ | 1 |
| | $= 24 - 14\sqrt{3}$ | 1 |
| (iii) | 378 | 1 |
| (iv) | $5 - 6\sqrt{10} + 18$ | 1 |
| | $= 23 - 6\sqrt{10}$ | 1 |
| (b) (i) | $\frac{5-2\sqrt{5}}{15}$ | 2 |
| | | (-1 if not simplified) |
| (ii) | $\frac{30-4\sqrt{10}-3\sqrt{10}+4}{45-8}$ | 2 (-1 for each mistake) |
| | $= \frac{34-7\sqrt{10}}{37}$ | 1 |
| (c) | $\frac{5+4\sqrt{5}+4+1}{\sqrt{5}+2}$ | 1 |
| | $= \frac{10+4\sqrt{5}}{\sqrt{5}+2} \times \frac{\sqrt{5}-2}{\sqrt{5}-2}$ | 1 |
| | $= 2\sqrt{5}$ | |
| | $\therefore b = 5$ | 1 |

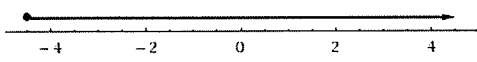
Question 4

- | | | |
|---------|------------------------|---|
| (a) (i) | $x = 9$ | 1 |
| (ii) | $3x + 15 - 2x + 2 = 6$ | 1 |
| | $x = -11$ | 1 |
| (iii) | $6x + 14 = 6 - x + 1$ | 1 |
| | $x = -1$ | 1 |
| (b) | 195°C | 3 |

Question 5

- (a) (i) $x - 3 = 5$ $-(x - 3) = 5$
 $x = 8$ $-x + 3 = 5$
 $x = -2$ **2**
- (ii) $2x + 6 = 3x - 1$ $-2x - 6 = 3x - 1$
 $x = 7$ $5x = -5$
 $x = -1$ **3**
- (b) $y = 9, x = -14$ **3**

Question 6

- (a) (i) $(x + 1)(x - 6)$ **1**
- (ii) $(3x - 2)(x - 4)$ **2**
- (iii) $x = \frac{4 \pm \sqrt{28}}{6}$
 $x = \frac{2 \pm \sqrt{7}}{3}$ **2**
- (b) (i) $2x + 8 = x - 5$ **1**
- (ii) $x = -13$ **1**
- (c) $3 \leq 12 + 2x$ **1**
 $-2x \leq 9$ **1**
 $x \geq -\frac{9}{2}$ **1**
- 
- (d) $(x - 6)(x + 5) < 0$
 $-5 < x < 6$ **2**
- (e) $4x + 10 > 13$ $-(4x + 10) > 13$
 $x > 0.75$ $x < -5.75$ **2**